

James Hornsby

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EDUCATION

University of Houston

Bachelor of Science in Computer Science and Mathematics (Double Major)

GPA: 3.58/4.00

Houston, TX

Expected May 2028

Computer Organization and Architecture

University of Houston

Fall 2025

- Studied the low-level operations of a computer.
- Established knowledge of hardware efficiency and performance.
- Grasped design concepts like datapaths and pipelining.

Introduction to Programming

University of Houston

Spring 2025

- Mastered basic object-oriented programming (OOP) principles through hands-on application.
- Grew a proficiency with programming concepts like inheritance and polymorphism.
- Engineered and developed a turn based RPG.

Introduction to Computer Science and Programming

University of Houston

Fall 2024

- Developed a foundational understanding of Computer Science.
- Internalized fundamental computer science concepts and grammatical thinking.

TECHNICAL SKILLS

Languages: C++, Python, HTML/CSS

Frameworks: Godot, Django, Flask, **FastAPI**, Playwright, **PyTorch**, Selenium

Graphics: **OpenGL**, **GLSL**, ASSIMP, GLEW, GLFW, GLM, SFML

Developer Tools: Android Studio, **Docker**, **Github**, Git, Visual Studio Code, Visual Studio, venv, WSL

Libraries: Beautiful Soup, LayoutParser, **OpenCV**, Pillow, **Pytesseract**, YOLO

EXPERIENCE

CodeCoogs Team Lead

University of Houston

Fall 2025 – Spring 2026

Houston, TX

- Lead a team of 7 students through the development of a **computer vision** application.
- Custom tuned **convolutional neural networks** for page parsing.
- Converted imaged books to flowed text with Pytesseract **OCR**.
- Utilized a simplistic **Agile** project management approach.

PROJECTS

Media Digitizer | *Docker, GitHub, OpenCV, Python, PyTorch, PyTesseract, YOLO*

August 2025 – Present

- Designed and built a **synthetic dataset** generator.
- Fine tuned **YOLOv8** computer vision model for page element classification.
- Prototyped an android application for Tensorflow Lite Inference through **Google ML Kit**.

CourtBooker | *Python, Playwright, GitHub*

August 2025

- Developed a tool to quickly reserve place at local tennis court.
- Utilized libraries like Time and Scheduler to allow delayed tasks like booking when availability opens.

Text Turn Based RPG | *C++*

May 2025

- Built a text battle RPG in C++ to demonstrate proficiency with numerous programming concepts.
- Implemented 8 classes to create a well organized and efficient design.
- Effectively handled memory to deliver a safe, leak free application.

Attendance Taker | *Python, requests*

June 2023

- Designed and scripted a Discord bot for a server of over 50.
- Implemented functionality to record event attendance and send reminders.